Statistical Techniques

Linear Regression

Vassilis Markos, Mediterranean College

Week 05

Contents

1 A Simple Example

A Simple Example

Good (?) Ol' Cars...

Consider the following simple car dataset about some 1973-74 models:

```
1 head(mtcars)
```

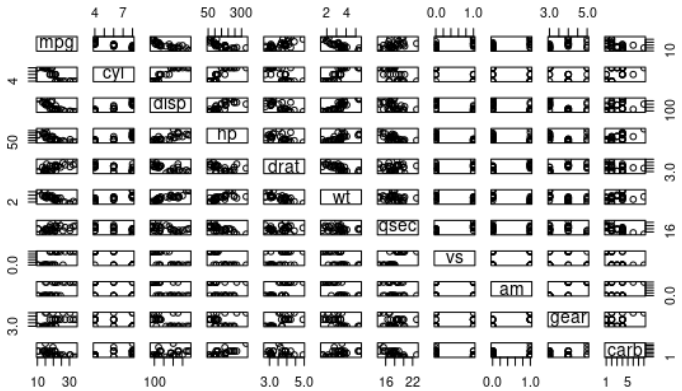
	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
Mazda RX4	21.0	6	160	110	3.90	2.620	16.46	0	1	4	4
Mazda RX4 Wag	21.0	6	160	110	3.90	2.875	17.02	0	1	4	4
Datsun 710	22.8	4	108	93	3.85	2.320	18.61	1	1	4	1
Hornet 4 Drive	21.4	6	258	110	3.08	3.215	19.44	1	0	3	1
Hornet Sportabout	18.7	8	360	175	3.15	3.440	17.02	0	0	3	2
Valiant	18.1	6	225	105	2.76	3.460	20.22	1	0	3	1

What Affects `mpg`?

- One valuable question might be which variables are correlated with which others and, more valuably, which ones predict others.
- One easy way to quick-check what goes on is to plot scatter plots of all data pairwise.
- The following command plots all such scatter plots twice (for X, Y interchangeably) for `mpg` against all other dataset variables.

```
1 pairs(mpg ~ ., data = mtcars)
```

What Affects `mpg`?

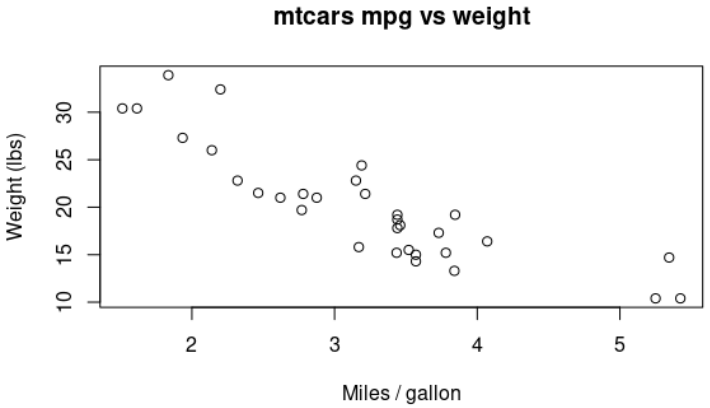


Okay, So...What?

- Take some time to meditate on the previous image...
- Since the plot matrix is symmetric, focus either on the lower or the upper triangular part.
- One correlation that we might consider exploring further should be between `mpg` and `wt` (i.e., miles per gallon and weight in lbs).
- So, we can plot this one scatter plot separately:

```
plot(mtcars$wt, mtcars$mpg, xlab = 'Miles / gallon', ylab = '
    Weight (lbs)', title(main='mtcars mpg vs weight'))
```

Okay, So…What?


mtcars mpg vs weight
Weight (lbs)
10
15
20
25
30
2
3
4
5
Miles / gallon

Among the Fogs of `mtcars`...

- ...a linear relationship between mpg and wt appears.
- So, how can we make sure that there actually is one?
- Here comes linear regression! A straight line has the shape of:

$$y = \beta_0 + \beta_1 x.$$

- The thing in our case is that, evidently, the points in the previous plot do not form a straight line. We can capture this deviation from a purely straight line by introducing an error term:

$$y = \beta_0 + \beta_1 x + \varepsilon.$$

Survival of the Fittest

- Okay, but, this uncertainty introduces an infinity of different choices for β_0, β_1, i.e., an infinity of candidate straight lines.
- Which one to choose? (Any ideas?)

Survival of the Fittest

- Okay, but, this uncertainty introduces an infinity of different choices for β_0, β_1, i.e., an infinity of candidate straight lines.
- Which one to choose? (Any ideas?)
- One criterion would be to minimise total error on each sample observation, i.e., for each estimated value of our model:

$$\hat{y}_i = \beta_0 + \beta_1 x_i, \quad i = 1, \ldots, n,$$

we should compute and minimise the corresponding errors:

$$\varepsilon_i = \hat{y}_i - y_i, \quad i = 1, \ldots, n.$$

But, What to Minimise?

- So, one approach here would be to minimise the total error:

$$SE = \sum_{i=1}^{n} \varepsilon_i.$$

- Can you see the pitfall here?

But, What to Minimise?

- So, one approach here would be to minimise the total error:

$$SE = \sum_{i=1}^{n} \varepsilon_i.$$

- Can you see the pitfall here?
- While summing all errors, positive and negative values might cancel out, resulting to misleading results.

But, What to Minimise?

- Another alternative would be to drop sign, i.e., compute and minimise:

$$SAE = \sum_{i=1}^{n} |\varepsilon_i| \, .$$

- This resolves the previous issue of errors cancelling, but, mathematically, the absolute value function is inconvenient to work with (i.e., non-differentiable).
- So, we end up with:

$$\boxed{SSE = \sum_{i=1}^{n} \varepsilon_i^2 .}$$

The Linear Regression Model

- Trying to minimise SSE by common multivariate calculus techniques, we find that the optimal values for β_0, β_1 should be:

$$\hat{\beta}_0 = \overline{y} - \hat{\beta}\overline{x}$$
$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_i - \overline{x})(y_i - \overline{y})}{\sum_{i=1}^{n}(x_i - \overline{x})^2}$$

- The standard error of the above model is defined as the square root of s^2, where:

$$s^2 = \frac{SSE}{n-1}.$$

Application on `mtcars`

```r
lg_model <- lm(wt~mpg, data=mtcars)
summary(lg_model)
```

```
Call:
lm(formula = mpg ~ wt, data = mtcars)

Residuals:
Min      1Q  Median      3Q     Max
-4.5432 -2.3647 -0.1252  1.4096  6.8727

Coefficients:
Estimate Std. Error t value Pr(>|t|)
(Intercept)  37.2851     1.8776  19.858  < 2e-16 ***
wt           -5.3445     0.5591  -9.559 1.29e-10 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 3.046 on 30 degrees of freedom
Multiple R-squared:  0.7528,    Adjusted R-squared:  0.7446
F-statistic: 91.38 on 1 and 30 DF,  p-value: 1.294e-10
```

Why R^2 and p-values?

- As you might observe, the previous slide contains some statistical terms that ring a bell…Can you suspect why?

Why R^2 and p-values?

- As you might observe, the previous slide contains some statistical terms that ring a bell...Can you suspect why?
- In order to test the model's fitness, we perform a hypothesis testing, where we assume that the errors are normally distributed, with a mean value of zero, and use as a null hypothesis:

$$H_0 = \beta_1 = 0.$$

- That is, under the assumption that errors are normally distributed around zero, we assume that there is no linear relationship between our two variables.

Why R^2 and p-values?

- Thus, we can use all our known hypothesis testing machinery to test whether or not a linear model is properly explaining the data we have at hand.
- For example, as with ANOVA, we have R^2 explain the percentage of variance in data captured by our model, since:

$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}.$$

- Recall: SSE is the errors sum of squares, SSR is the regression sum of squares and SST is the total sum of squares.

(Reminder) Errors

As a brief reminder, with our notation so far:

$$SST = SSR + SSE$$

$$SST = \sum_{i=1}^{n} (\hat{y}_i - y_i)^2$$

$$SSR = \sum_{i=1}^{n} (\hat{y}_i - \overline{y})^2$$

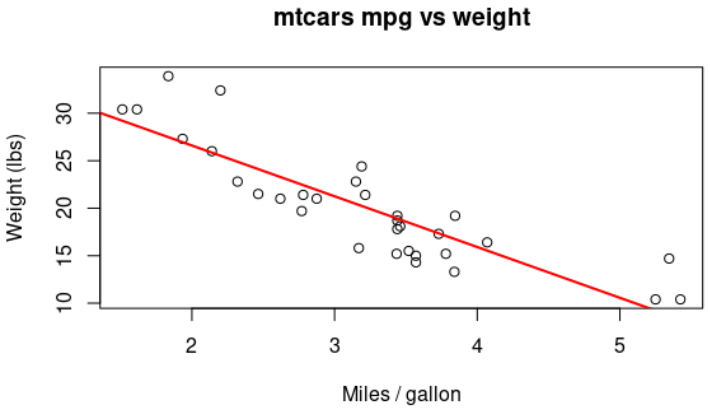
$$SSE = \sum_{i=1}^{n} (y_i - \overline{y})^2$$

So, How Does Our Model Look Like?

- To plot our model over the scattered data points, we have to plot both one on top of the other.
- Remember to plot your variables the same way they appear in your model, i.e., y against x, otherwise, you might get a non-sensical plot!

```
plot(mtcars$wt, mtcars$mpg, xlab = 'Miles / gallon', ylab = '
    Weight (lbs)', title(main='mtcars mpg vs weight'))
abline(lg_model, col = 'red', lwd = 2)
```

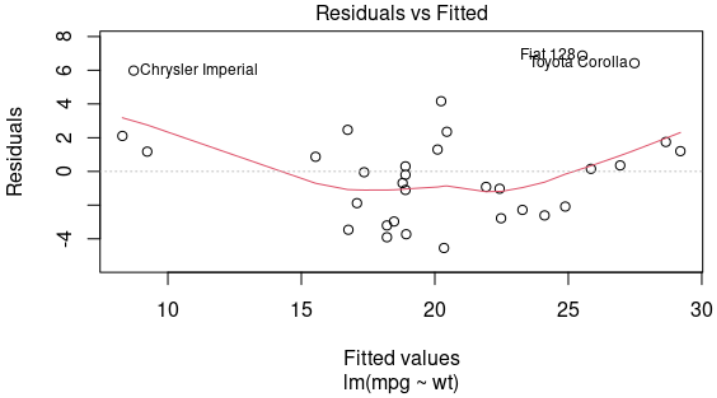
Our Model!


mtcars mpg vs weight
Weight (lbs)
10
15
20
25
30
2
3
4
5
Miles / gallon

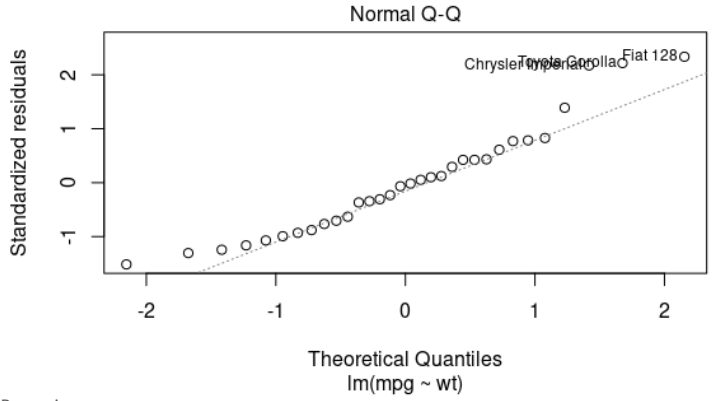
Further Model Evaluation

- **Residuals vs Fitted** plots, which plot the predicted values versus the corresponding errors. Ideally, it should look like randomly arranged points with no significant patterns.
- **Normal Q-Q** plots, which show standardised residuals vs their theoretical values. Ideally, all points should like on the diagonal.
- **Scale-location** plots, which plots predicted against square root standardised errors.
- **Residuals-Leverage** plots, which identify points that might affect the regression line a lot.

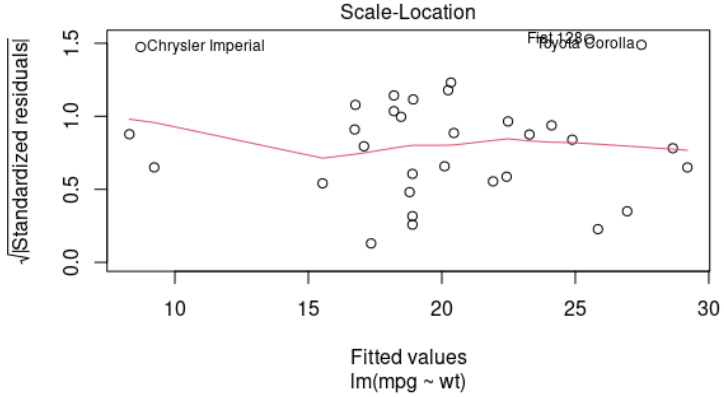
Residuals vs Fitted



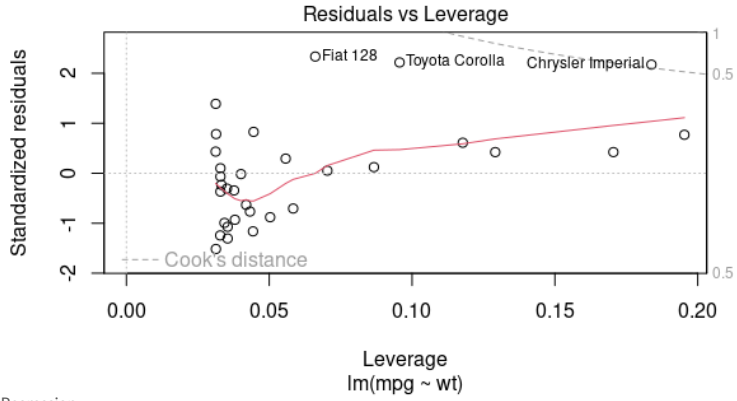
Normal Q-Q


Normal Q-Q
Standardized residuals
Theoretical Quantiles
lm(mpg ~ wt)
Chrysler Imperial
Toyota Corolla
Fiat 128
-2
-1
0
1
2

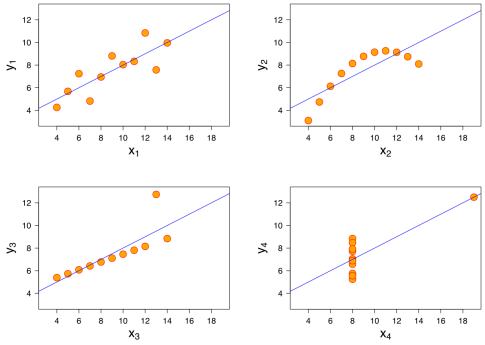
Scale-Location



Residuals vs Leverage



Don’t Rely Just on Regression!



Source: https://en.wikipedia.org/wiki/Linear_regression#Least-squares_estimation_and_related_techniques